Paper Study

High-temperature superconductivity in sulfur hydride evidenced by alternating-current magnetic susceptibility

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PHYSICS

High-temperature superconductivity in sulfur hydride evidenced by alternating-current magnetic susceptibility

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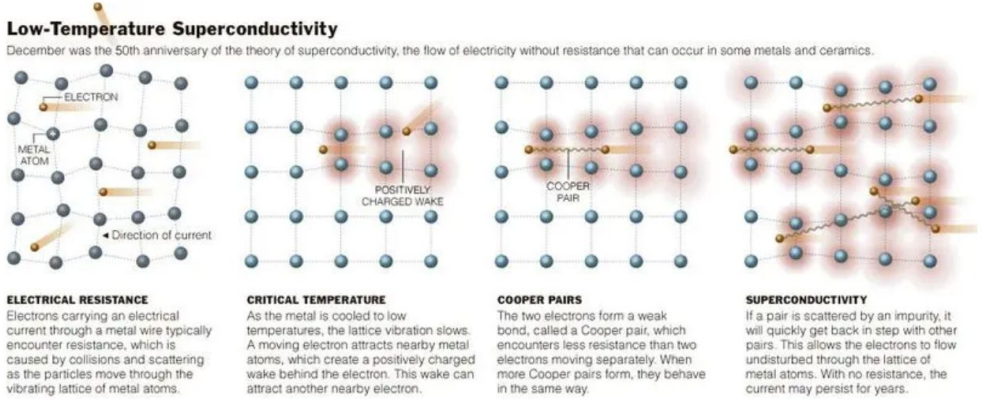
Contents

1. The material investigated in this study: hydrogen sulfide
2. Identification of the superconducting state – Meissner effect
3. Setup for AC magnetic susceptibility measurement
4. Sample preparation and first experiment
5. Second experiment and comparison
6. Reason for the decrease of Tc above 150 GPa
7. Discrepancies between measurement methods

The material investigated in this study: hydrogen sulfide

High Debye temperatures:

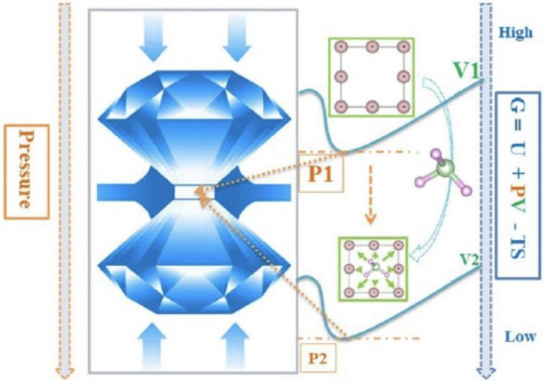
- Indicate **high-frequency phonons**, usually from light elements and **strong bonds**.
- In phonon-mediated superconductors, these phonons enhance electron–phonon coupling, **polarizing the lattice** and producing an effective delayed attraction that overcomes Coulomb repulsion, forming Cooper pairs (BCS theory).
- Below Tc, the superconducting gap opens, **suppressing electron–phonon scattering** and enabling zero resistance.



The material investigated in this study: hydrogen sulfide

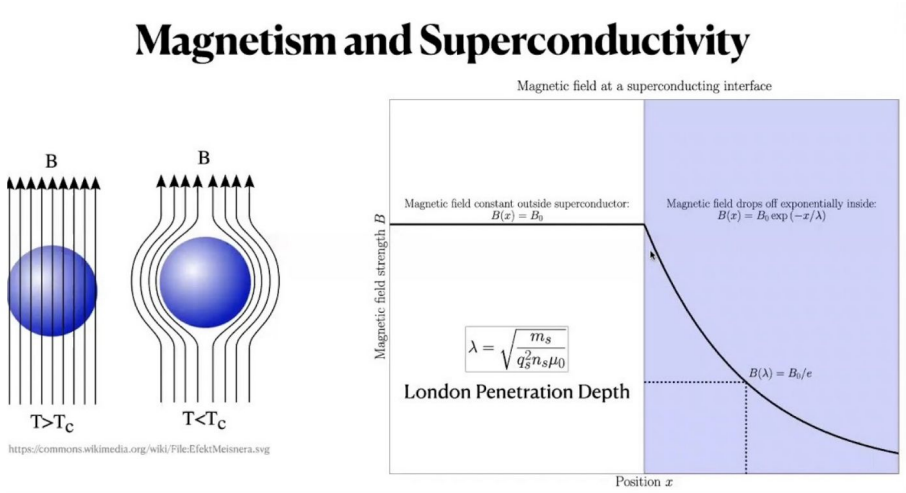
Chemical precompression:

- Heavy atoms in hydrogen-rich compounds (e.g., sulfur in H_2S) provide **chemical precompression**, lowering the required external pressure.
- Under high pressure, H_2S converts to H_3S, introducing hydrogen-derived states near the Fermi level, enhancing **electron–phonon coupling** and enabling high-temperature superconductivity.



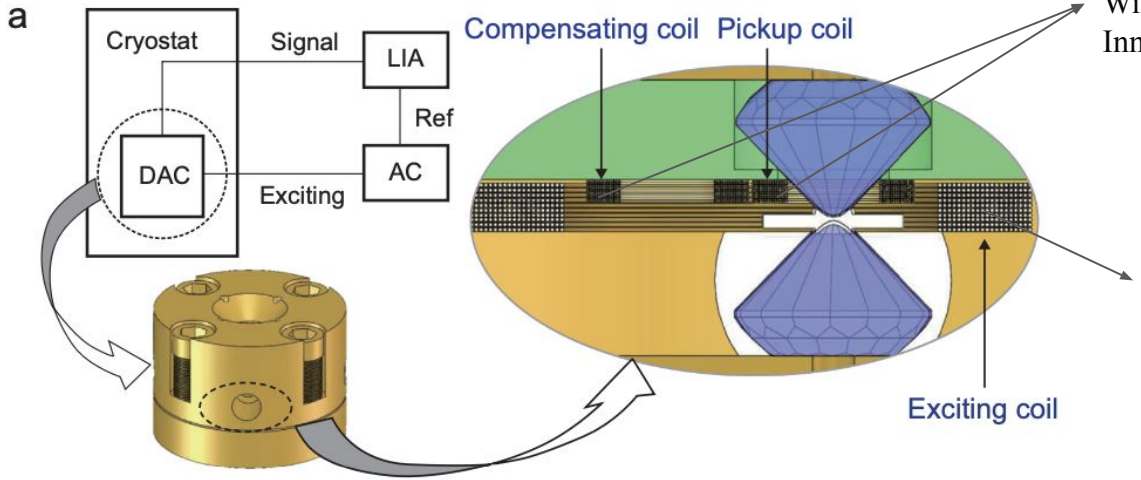
Identification of the superconducting state – Meissner effect

- **Below the critical temperature Tc**, a superconductor in a weak external magnetic field allows **the field to penetrate only slightly**, decaying to 1/e of its surface value over the London penetration depth.
- The superconducting transition is identified by monitoring **changes in magnetic susceptibility**, which reflect the emergence of **diamagnetism**.



Setup for AC magnetic susceptibility measurement

- **Compensating coil: oppositely connected to the pickup coil** to **cancel the constant susceptibility signal** originating from the surrounding metallic components.
- **LIA:** retrieve the amplitude and phase of small signals embedded in noise

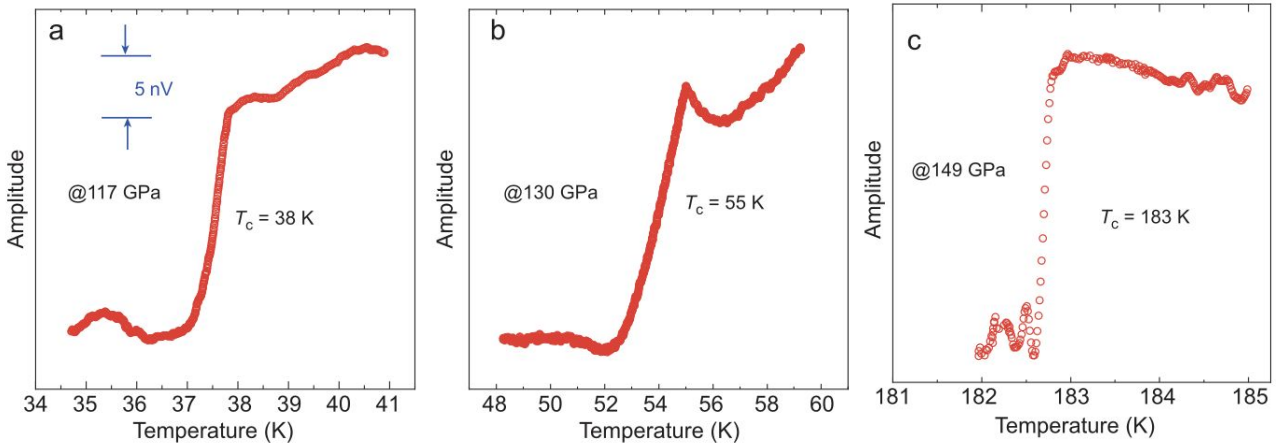


Number of turns: 180
Wire diameter: 20 μm (insulated copper wire)
Inner diameter: 2 mm

Number of turns: 300
Wire diameter: 38 μm (insulated copper wire)
Inner diameter: 9.4 mm
RMS current: 10 mA
Magnetic field: 0.75×10^{-4} T

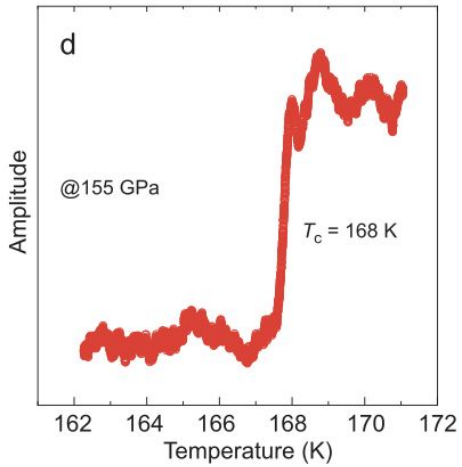
Sample preparation and first experiment

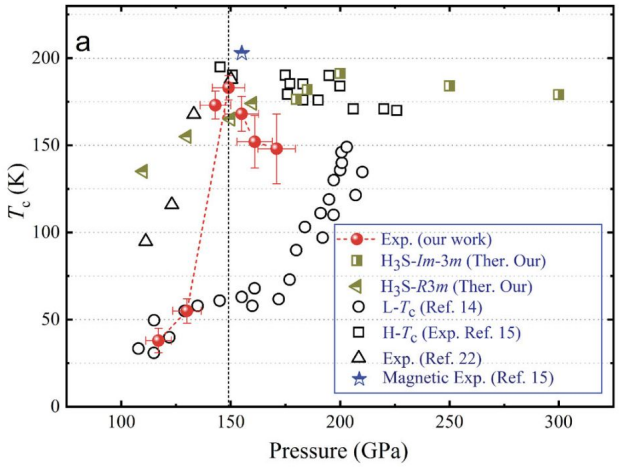
- culet size: 100 μm, preindented gasket: 20 μm, hole diameter: 60 μm
- In an argon glove box, H_2S was condensed into the gasket hole cooled in liquid nitrogen, and **the sample was subsequently compressed to at least 100 GPa at T < 100 K** to avoid possible decomposition.
- In the first run of the experiment, **superconducting transitions** were observed at **117 GPa and 130 GPa**, with **Tc values of 38 K and 55 K**, respectively.
- At 149 GPa, the sample was held at 300 K for seven days to allow the formation of the **H_3S phase**, resulting in a **Tc of 183 K**.



Second experiment and comparison

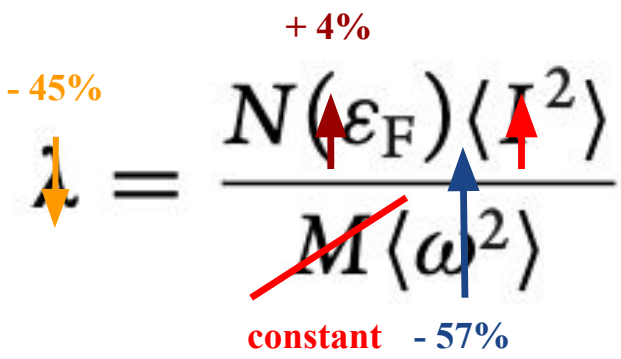
- At 155 GPa, Tc=168 K; further compression lowers Tc
- **Above 150 GPa**, the diameter of the diamond anvil culets is reduced, leading to a **decreased signal-to-noise ratio**.
- **Below 140 GPa**, **multiple hydrogen-rich H–S phases coexist** in the sample in addition to H_3S, and the low-Tc data can be reproduced by theoretical calculations based on stoichiometry.

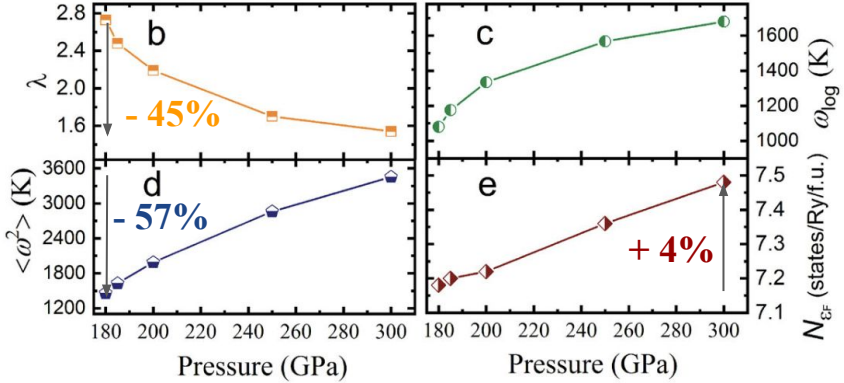




Reason for the decrease of Tc above 150 GPa

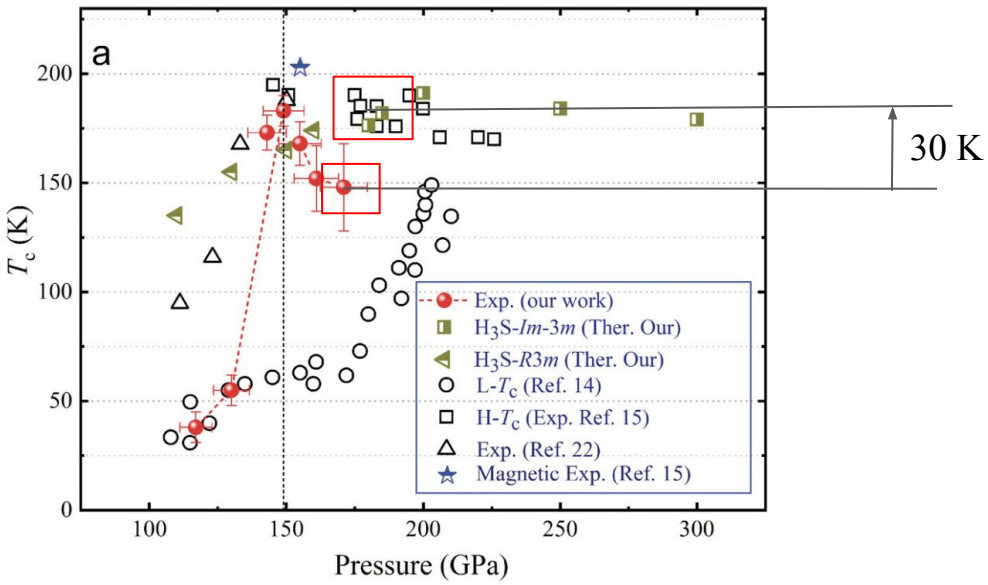
- Above 150 GPa, the decrease in Tc is consistent with the formation of the **Im-3m phase** and is primarily dominated by the **electron–phonon coupling constant λ**.
- At higher pressures, N(εF) and ⟨I²⟩ both increase. Consequently, **the reduction in λ** can only be attributed to the **increase in phonon frequencies** arising from pressure-induced **phonon hardening**.





Discrepancies between measurement methods

- The critical temperature determined from **electrical resistance measurements** is consistently **higher** than that obtained from **magnetic susceptibility**, with a difference of about **30 K** at 170 GPa.
- Resistance measurements are sensitive to **small superconducting fractions** forming **percolation paths**, whereas magnetic measurements probe only the **bulk superconducting fraction**.



Reference

1. Xiaoli Huang, Xin Wang, Defang Duan, Bertil Sundqvist, Xin Li, Yanping Huang, Hongyu Yu, Fangfei Li, Qiang Zhou, Bingbing Liu, Tian Cui, High-temperature superconductivity in sulfur hydride evidenced by alternating-current magnetic susceptibility, *National Science Review*, Volume 6, Issue 4, July 2019, Pages 713–718, https://doi.org/10.1093/nsr/nwz061
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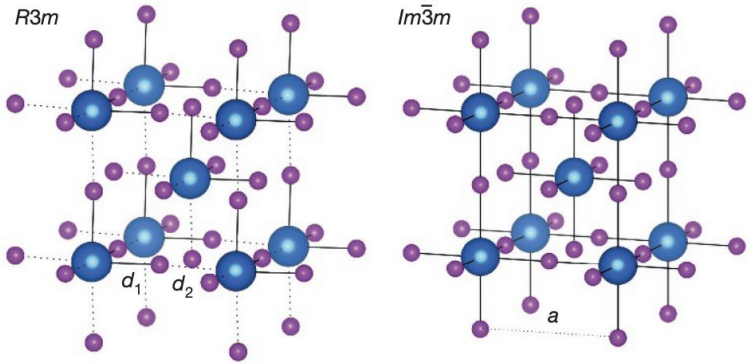
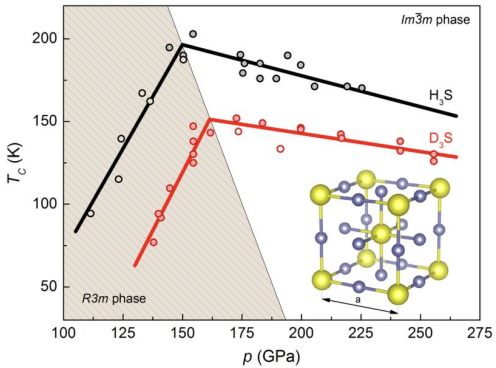
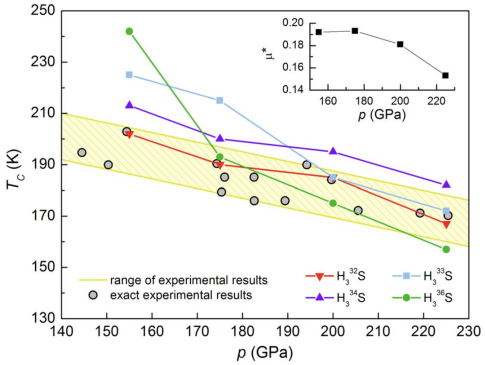
Relevant literature data and underlying principles

- H_3S reaches a maximum Tc of ~203 K at ~155 GPa.
- In the **Im3m structure**, **quantum symmetrization** reduces the pressure required for hydrogen-bond symmetrization, and this phase dominates **the pressure range where high Tc is observed**.
- Replacing sulfur with the heavier isotope **Sulfur-36 is predicted to raise Tc to ~242 K**, and this **anomalous isotope effect** indicates strong phonon coupling, supporting H_3S as a **phonon-mediated superconductor.**